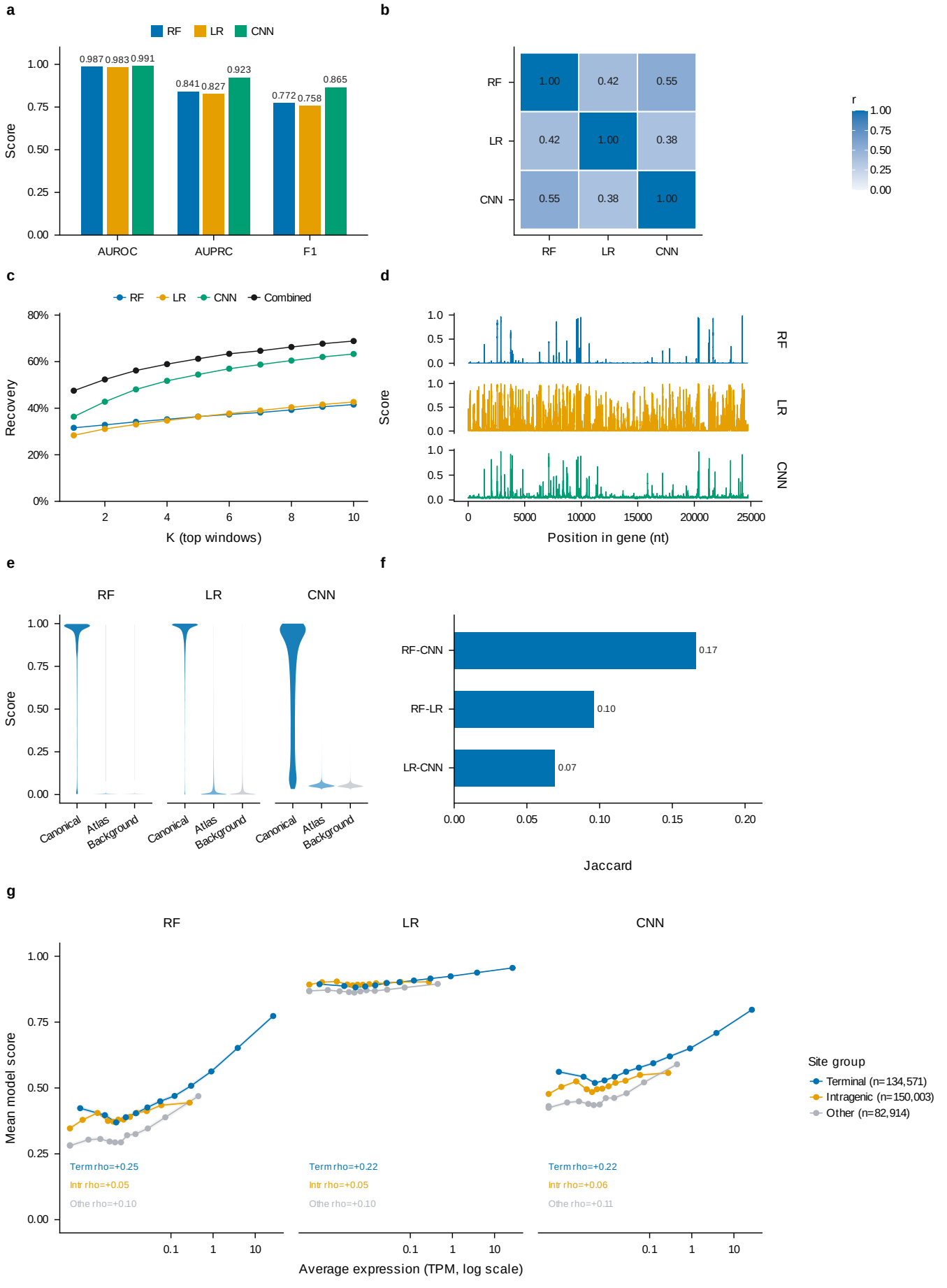

a
RF
LR
CNN
Score
1.00
0.75
0.50
0.25
0.00
0.987
0.983
0.991
0.841
0.827
0.923
0.772
0.758
0.865
AUROC
AUPRC
F1
b
RF
LR
CNN
1.00
0.42
0.55
0.42
1.00
0.38
0.55
0.38
1.00
r
1.00
0.75
0.50
0.25
0.00
c
RF
LR
CNN
Combined
Recovery
80%
60%
40%
20%
0%
2
4
6
8
10
K (top windows)
d
Score
1.0
0.5
0.0
RF
LR
CNN
0
5000
10000
15000
20000
25000
Position in gene (nt)
e
RF
LR
CNN
Score
1.00
0.75
0.50
0.25
0.00
Canonical
Atlas
Background
f
RF-CNN
RF-LR
LR-CNN
0.17
0.10
0.07
0.00
0.05
0.10
0.15
0.20
Jaccard
g
RF
LR
CNN
Mean model score
1.00
0.75
0.50
0.25
0.00
Term rho=+0.25
Intr rho=+0.05
Othe rho=+0.10
Term rho=+0.22
Intr rho=+0.05
Othe rho=+0.10
Term rho=+0.22
Intr rho=+0.06
Othe rho=+0.11
0.1
1
10
Average expression (TPM, log scale)
Site group
Terminal (n=134,571)
Intragenic (n=150,003)
Other (n=82,914)